IDEATION PHASE

Brainstorm & Idea Prioritization Template

Date	30 July 2026
Name	Kesireddy Padma
Project Name	Metro Ticket Generating System in ServiceNow
Maximum Marks	4 marks

Step-1: Individual Mobilization, Research, and Problem Identification

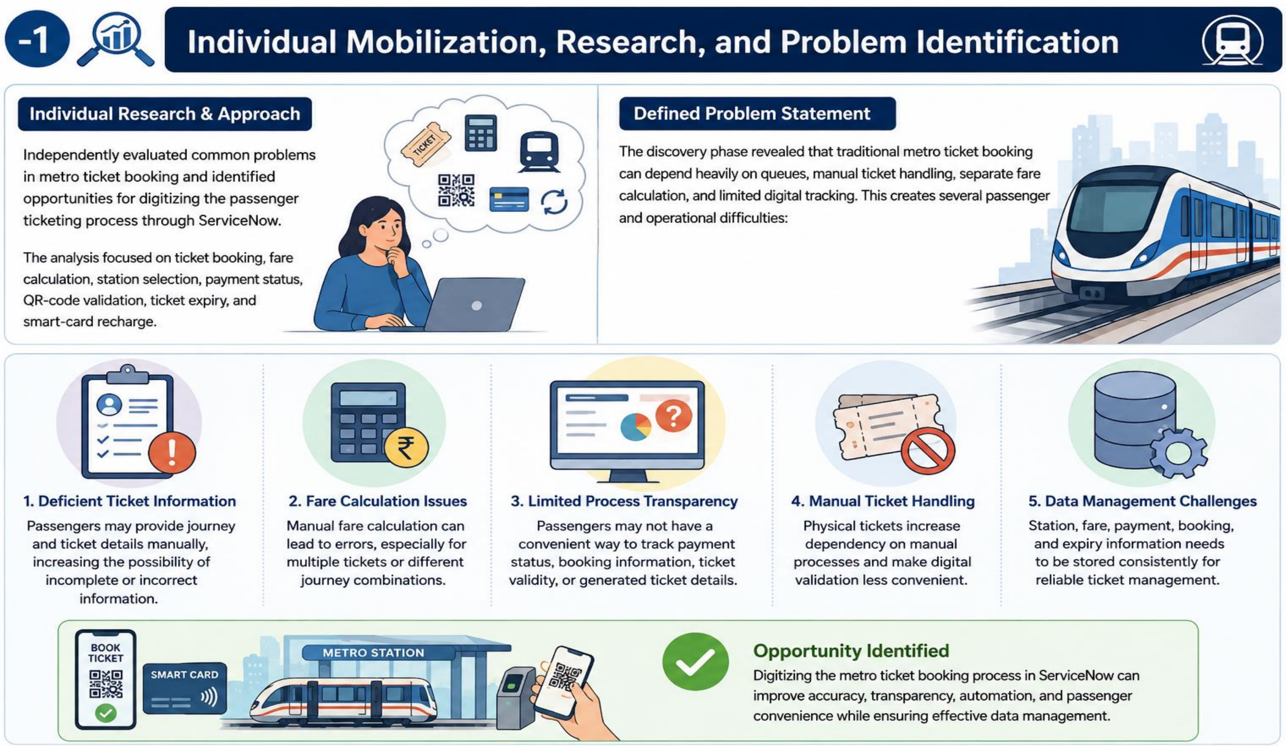
Individual Research & Approach

Independently evaluated common problems in metro ticket booking and identified opportunities for digitizing the passenger ticketing process through ServiceNow. The analysis focused on ticket booking, fare calculation, station selection, payment status, QR-code validation, ticket expiry, and smart-card recharge.

Defined Problem Statement

The discovery phase revealed that traditional metro ticket booking can depend heavily on queues, manual ticket handling, separate fare calculation, and limited digital tracking. This creates several passenger and operational difficulties:

- **Deficient Ticket Information**: Passengers may provide journey and ticket details manually, increasing the possibility of incomplete or incorrect information.
- **Fare Calculation Issues**: Manual fare calculation can lead to errors, especially for multiple tickets or different journey combinations.
- **Limited Process Transparency:** Passengers may not have a convenient way to track payment status, booking information, ticket validity, or generated ticket details.
- **Manual Ticket Handling:** Physical tickets increase dependency on manual processes and make digital validation less convenient.
- **Data Management Challenges:** Station, fare, payment, booking, and expiry information needs to be stored consistently for reliable ticket management.



Step-2: Taxonomic Categorization & Brainstorming Repository

Group A: Presentation Layer & Front-End Validation Protocols

• System Component 1 (Intelligent Service Catalog): Deploy a user-friendly Metro Ticket Service Catalog interface to capture source station, destination station, ticket type, number of tickets, and related passenger details.

• System Component 2 (Deterministic Field Constraints): Apply mandatory field validation so essential journey and ticket information is provided before submission.

• System Component 3 (Contextual UI Logic): Use Catalog Client Scripts and UI Policies to dynamically control field behavior and display relevant ticket-booking information based on passenger selections.

Group B: Ticket Processing & Lifecycle Automation Engine

• System Component 4 (Automated Fare Calculation): Calculate the applicable fare based on selected source, destination, and number of tickets.

• System Component 5 (Automated Ticket Processing): Use Flow Designer to automate processing of submitted ticket requests and related record updates.

• System Component 6 (Ticket Status Controls): Maintain payment status, booking status, and ticket expiry information so the ticket lifecycle can be tracked correctly.

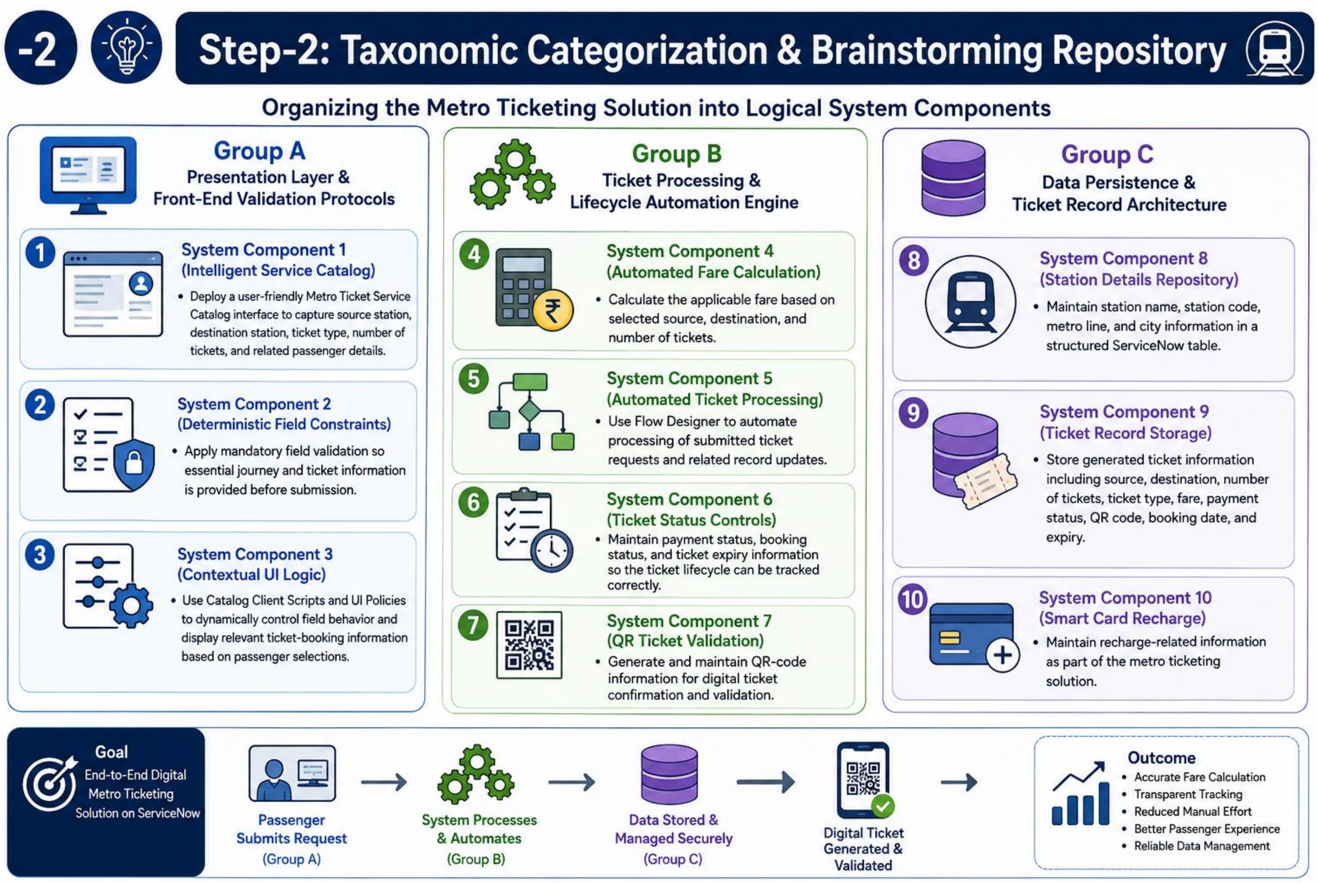
• System Component 7 (QR Ticket Validation): Generate and maintain QR-code information for digital ticket confirmation and validation.

Group C: Data Persistence & Ticket Record Architecture

• System Component 8 (Station Details Repository): Maintain station name, station code, metro line, and city information in a structured ServiceNow table.

• System Component 9 (Ticket Record Storage): Store generated ticket information including source, destination, number of tickets, ticket type, fare, payment status, QR code, booking date, and expiry.

• System Component 10 (Smart Card Recharge): Maintain recharge-related information as part of the metro ticketing solution.



Step-3: Feasibility Analysis & Strategic Prioritization

The proposed Metro Ticket Generating System components were evaluated according to implementation effort and expected operational impact. The following prioritization identifies the features that provide the strongest value to passengers and the ticketing process.

Architectural Core Concept	Technical Complexity (Effort)	Operational Return (Impact)	Strategic Execution & Action

Dynamic Catalog UI Policies	Low	High	High Priority: Adopted to provide simple, context-aware ticket-booking fields and validation.
Automated Fare Calculation	Medium	High	High Priority: Adopted to reduce manual fare calculation and improve booking accuracy.
Flow Designer Ticket Automation	Medium	High	High Priority: Adopted as the primary automation engine for ticket processing.
QR Code Ticket Validation	Medium	High	High Priority: Adopted for convenient digital ticket confirmation and validation.
Structured Station & Ticket Records	Medium	High	High Priority: Adopted to maintain clean, searchable, reliable ticket information.
Basic Notifications	Low	Medium	Supporting Priority: Retained for booking and ticket-status communication.
Smart Card Recharge	Medium	Medium	Secondary Priority: Included as an additional capability for future expansion.

Final Prioritization: Dynamic UI validation, automated fare calculation, Flow Designer automation, QR-code validation, and structured ticket/station records were identified as the core high-priority features because they directly improve passenger convenience, accuracy, automation, and ticket traceability.

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Structured Station & Ticket Records	Medium	High	HIGH PRIORITY	Adopted to maintain clean, searchable, reliable ticket information.
Basic Notifications	Low	Medium	SUPPORTING PRIORITY	Retained for booking and ticket-status communication.
Smart Card Recharge	Medium	Medium	SECONDARY PRIORITY	Included as an additional capability for future expansion.

FINAL PRIORITIZATION:

Dynamic UI validation, automated fare calculation, Flow Designer automation, QR-code validation, and structured ticket/station records were identified as the core high-priority features because they directly improve passenger convenience, accuracy, automation, and ticket traceability.

- Dynamic UI Validation
- Automated Fare Calculation
- Flow Designer Automation
- QR Code Validation
- Structured Records